

BRAD BURCH

bradburch.jobs@gmail.com | (574) 835-6870 | San Francisco, CA | linkedin.com/in/burchbrad/ | bradburch.github.io

SUMMARY

Software engineer with 7 years shipping production full-stack systems and LLM agents (Claude, MCP, OAuth 2.1 + PKCE) on Cloudflare Workers, D1, KV, and R2 with TypeScript, Hono, and React 19, from schema design through ship and on-call behind a CI-gated Vitest and Playwright suite.

SKILLS

Languages: TypeScript, JavaScript, Python, Java, Scala, SQL

Frameworks & Runtimes: React 19, React Router 8, Node.js, Hono, Spring, Play, Cloudflare Workers

Infrastructure: Cloudflare (Workers, D1, KV, R2, Workflows), AWS (DynamoDB, SQS), PostgreSQL, MSSQL, SQLite, Docker, REST/JSON APIs, schema migrations, idempotency

AI / LLM: Claude (Haiku, Sonnet), Vercel AI SDK, Model Context Protocol (MCP) servers, agentic tool loops, eval harnesses, prompt-injection mitigation, streaming, AI-assisted development (Claude Code, Cursor, AI coding agents)

Auth & Security: OAuth 2.1, PKCE (RFC 7636), JWT (HS256), SAML 2.0, refresh token rotation, multi-tenant isolation, CSRF and XSS hardening

Testing & Delivery: Vitest, Playwright, GitHub Actions, CI/CD, Jenkins, structured JSON logging, production monitoring & alerting

EXPERIENCE

Software Engineer

Sept 2022 - Present

Pawsservation, San Francisco, CA (pawsservation.com) | sole engineer and operator

- Built a production TypeScript client portal and API on Cloudflare Workers (D1, KV, R2, Workflows) with no origin server, serving 30+ active clients, 60+ bookings/month, and 30+ invoices/month.
- Added change detection to a scheduled calendar sync, eliminating the source of 97% of daily database writes (56,424 of 58,121 against a 100,000/day platform cap).
- Replaced date-ordered payment matching with closest-first allocation, ending a defect class that cleared a \$600 stay with a \$40 payment and flagged a \$2,000 prepayment as delinquent.
- Rebuilt a calendar backfill blocked by a 50-subrequest platform ceiling into a durable Cloudflare Workflows fan-out with crash-safe cursor checkpointing and idempotent fan-in.
- Shipped a customer-facing Claude Haiku 4.5 booking agent (Vercel AI SDK, 15 tools) for booking, rescheduling, and cancellation, with two-phase confirmation on a single-use token per write.
- Implemented an MCP server with a hand-written OAuth 2.1 + PKCE flow (S256, single-use codes, rotating hashed refresh tokens), letting any spec-compliant client connect self-service.
- Isolated tenant data across config, pricing, capacity, and bookings by scoping every SQL query through a single business-id module.
- Hardened payment ingestion with SPF and DKIM sender verification, revocable server-side sessions, and RFC 5987 filename encoding against header injection.

Software Engineer

May 2022 - Aug 2022

Front, San Francisco, CA (role eliminated in company-wide layoff)

- Shipped a Node.js REST API enabling account-level email rules for multi-domain enterprise customers, previously configurable only per-inbox.
- Performed a security upgrade on the core JavaScript email library responsible for 200K+ outbound messages per month.

Software Engineer

May 2019 - Aug 2021

Total Brain, San Francisco, CA

- Shipped a Scala backend integration for Apple, Google, and Facebook authentication, contributing to a 60% lift in new sign-ups.
- Migrated user data from MSSQL to DynamoDB with a redesigned NoSQL schema, cutting average API response times by 50%; mentored junior engineers on backend patterns.
- Standardized SAML 2.0 and JWT SSO and built HubSpot and Box integrations, cutting enterprise onboarding from 2+ weeks to ~3 days.
- Built a Jenkins CI/CD pipeline and release automation, increasing deploy frequency and adding structured logs for observability.

Software Engineer

Jul 2017 - Aug 2018

Salesforce.com, San Francisco, CA

- Built automated testing frameworks on a developer-productivity team, including a persistent test-state repository (Java, PostgreSQL, EclipseLink).
- Built an OAuth2 security filter for microservice API access control with remote identity validation.
- Built a React management UI that internal engineers used to configure, launch, and inspect microservice test runs, cutting internal test setup time by 40%.
- Led delivery of a config builder and validator used by internal teams, scoping milestones and coordinating partner teams.

EDUCATION

DePauw University, Bachelor of Arts in Computer Science

2017